

MEDORA 7 E, ND, USA

WMO: 727520

Lat: 46.8946N

Lon: 103.3769W

Elev: 2771

StdP: 13.28

Time Zone: -7.00 (NAM)

Period: 04-19

WBAN: 94080

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-14.5	-9.6	-20.8	1.9	-14.4	-15.7	2.6	-9.8	18.2	30.6	16.5	27.6	5.7	N/A	0.565

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.2	92.1	65.0	88.4	64.1	85.0	63.6	69.5	82.5	67.7	81.2	65.8	79.7	7.2	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.5	104.3	76.0	63.1	96.0	73.8	61.0	88.8	71.1	35.4	82.0	33.6	81.2	32.1	79.7	81.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.7	14.7	13.1	DB	-21.4	98.0	4.0	4.1	-24.3	101.0	-26.7	103.4	-28.9	105.8	-31.9	108.8	(4)
(5)			WB	-22.0	73.7	3.9	3.6	-24.8	76.3	-27.1	78.5	-29.3	80.5	-32.1	83.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	43.5	19.0	17.9	31.5	41.9	52.8	63.1	71.0	68.5	59.1	44.3	31.8	19.2	(6)
	DBStd	21.62	14.39	14.19	13.06	10.15	8.51	6.80	6.02	6.44	9.00	9.80	12.07	13.95	(7)
	HDD50	4551	962	899	583	273	71	4	0	0	24	229	552	954	(8)
	HDD65	8307	1427	1319	1040	692	386	110	15	39	220	643	997	1419	(9)
	CDD50	2172	0	0	9	31	156	398	650	574	296	53	5	0	(10)
	CDD65	452	0	0	0	0	7	53	200	148	42	2	0	0	(11)
	CDH74	5343	0	0	2	20	121	582	2218	1753	606	41	0	0	(12)
	CDH80	2066	0	0	0	3	20	160	939	720	213	11	0	0	(13)
(14) Wind	WSAvg	6.9	7.5	6.8	7.4	7.7	7.2	6.6	6.1	6.1	6.4	6.8	6.8	6.8	(14)
(15) Precipitation	PrecAvg	15.3	0.5	0.4	0.7	1.4	2.3	3.2	2.2	1.2	1.3	1.2	0.6	0.4	(15)
	PrecMax	20.0	1.3	1.6	2.0	3.6	4.5	5.6	5.9	3.0	3.0	4.2	2.9	1.1	(16)
	PrecMin	10.1	0.1	0.0	0.2	0.4	0.7	1.1	0.6	0.1	0.2	0.2	0.0	0.0	(17)
	PrecStd	2.4	0.3	0.3	0.5	0.8	1.3	1.2	1.3	0.6	0.8	1.1	0.6	0.3	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	51.1	53.8	70.4	76.6	82.7	90.2	98.1	96.2	91.5	78.8	66.5	51.2	(19)
		MCWB	39.9	42.1	48.8	53.5	58.7	65.8	65.6	63.4	64.4	56.5	49.1	40.5	(20)
	2%	DB	44.7	46.3	62.4	68.8	77.3	84.5	92.8	91.4	86.0	71.1	58.7	45.0	(21)
		MCWB	37.1	38.4	45.7	50.0	58.6	64.4	65.7	64.3	61.6	54.1	45.5	36.8	(22)
	5%	DB	40.8	41.5	56.4	63.8	73.2	80.3	88.9	87.7	81.6	64.9	53.0	41.0	(23)
		MCWB	34.8	35.2	43.2	47.5	56.2	62.8	65.3	63.1	59.2	50.4	41.9	34.3	(24)
	10%	DB	37.4	37.1	50.2	59.3	68.9	76.8	85.5	83.8	76.2	59.9	48.3	37.5	(25)
		MCWB	32.6	32.5	40.5	45.2	54.4	61.0	65.6	62.7	58.2	48.0	39.5	32.1	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	40.3	43.9	50.9	55.0	64.7	71.3	74.7	70.1	67.3	58.9	49.5	41.8	(27)
		MCDB	49.1	52.9	68.0	75.4	74.4	83.6	84.0	83.9	80.5	74.4	65.6	50.9	(28)
	2%	WB	37.1	39.3	47.1	50.6	61.4	67.8	70.8	68.0	64.3	56.2	45.5	37.6	(29)
		MCDB	44.1	47.2	61.9	65.2	72.4	79.7	81.5	81.6	78.6	69.2	57.4	44.4	(30)
	5%	WB	34.7	35.4	44.6	48.0	58.9	64.8	68.6	66.1	61.8	52.1	42.1	34.9	(31)
		MCDB	39.9	41.2	57.5	61.1	69.5	77.3	82.2	80.5	76.9	63.2	51.1	40.9	(32)
	10%	WB	32.6	33.1	41.5	45.4	56.1	62.7	66.6	64.4	59.3	48.9	39.1	32.4	(33)
		MCDB	37.0	37.5	51.2	57.3	66.2	74.6	80.7	78.6	72.3	58.9	46.4	37.4	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	17.1	17.0	18.8	20.5	20.0	20.6	23.2	23.9	23.0	19.1	18.2	15.7	(35)
		MCDBR	18.5	21.9	28.4	29.2	27.2	26.8	29.2	30.4	31.4	28.7	25.7	20.1	(36)
	5% WB	MCWBR	11.7	14.1	15.7	15.6	12.7	12.9	10.3	10.9	12.8	14.4	14.9	12.8	(37)
		MCWBR	17.6	19.9	28.1	28.4	22.1	24.0	24.4	25.5	27.3	26.5	24.4	19.5	(38)
(40) Clear-Sky Solar Irradiance	taub		0.249	0.256	0.273	0.310	0.330	0.343	0.353	0.360	0.319	0.290	0.260	0.245	(40)
		taud	2.517	2.538	2.550	2.447	2.454	2.448	2.451	2.400	2.495	2.560	2.535	2.523	(41)
	Ebn at Noon		278	298	307	301	297	292	287	280	285	280	270	266	(42)
		Edh at Noon	19	23	27	33	34	35	34	34	29	23	19	17	(43)
(44) All-Sky Solar Radiation	RadAvg		519	831	1195	1555	1841	2077	2195	1809	1368	868	547	393	(44)
	RadStd		41	85	91	92	165	156	89	120	103	95	65	20	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (11 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page